UNDERSTANDING THE DOM & DOM EVENTS IN JAVASCRIPT

JavaScript interacts with web pages using the Document Object Model (DOM). Today, let's break down:

✅ What is the DOM?
✅ How to select & manipulate elements?
✅ Handling DOM Events

What is the DOM?

The DOM (Document Object Model) represents a web page as a tree-like structure where each element is a node. JavaScript allows us to access and modify these elements dynamically.

```html
<!DOCTYPE html>
<html>
  <head>
    <title>DOM Example</title>
  </head>
  <body>
    <h1 id="title">Hello, DOM!</h1>
    <p class="content">This is a paragraph.</p>
    <button onclick="changeText()">Click Me</button>

    <script>
      function changeText() {
        document.getElementById("title").innerText = "DOM Updated!";
      }
    </script>
  </body>
</html>
```

🔹 When the button is clicked, the DOM updates dynamically using JavaScript.

2. Selecting Elements in the DOM

JavaScript provides multiple ways to select elements from the DOM:

✅ Select by ID (getElementById)

```
let title =
document.getElementById("title");
console.log(title.innerText); //
"Hello, DOM!"
```

✅ Select by Class (getElementsByClassName)

```
let paragraphs =
document.getElementsByClassName("content
");
console.log(paragraphs[0].innerText); //
"This is a paragraph."
```

✅ Select by Tag(getElementsByTagName)

```
let allParagraphs =
document.getElementsByTagName("p");
console.log(allParagraphs.length); //
Number of <p> elements
```

✅ Modern Approach: querySelector &querySelectorAll

```
let title =
document.querySelector("#title"); //
Selects by ID
let allParagraphs =
document.querySelectorAll(".content"); //
Selects all matching elements
```

🔹 querySelector (single element) & querySelectorAll (all matching elements) are recommended over older methods.

3. Modifying the DOM

✅ Change Text Content

```
document.getElementById("title").innerText
= "New Title!";
```

✅ Change HTML Content

```
document.getElementById("title").innerHTML
= "<span>Updated!</span>";
```

✅ Change CSS Styles

```
document.getElementById("title").style.color = "blue";
```

✅ Add & Remove Classes

```
document.getElementById("title").classList.add("highlight"); // Add class
document.getElementById("title").classList.remove("highlight"); // Remove class
```

4. Handling DOM Events

Events are actions that occur in the browser, like:

✅ Clicking a button (click)

✅ Typing in an input field (input)

✅ Submitting a form (submit)

✅ Handling Click Events

```
document.getElementById("title").addEventLis
tener("click", function () {
  alert("Title Clicked!");
});
```

✅ Handling Input Events

```
document.getElementById("myInput").
addEventListener("input", function
(event) {
  console.log(event.target.value);
// Logs input value
});
```

✅ Handling Form Submission

```
document.getElementById("myForm").addEvent
Listener("submit", function (event) {
  event.preventDefault(); // Prevent page
refresh
  console.log("Form Submitted!");
});
```

✅ Removing an Event Listener

```
function sayHello() {
  console.log("Hello!");
}

document.getElementById("btn").addEventLi
stener("click", sayHello);
document.getElementById("btn").removeEven
tListener("click", sayHello);
```